Second and Higher Order Derivatives

Q1. First and Second Derivatives. Find the first and second derivatives of the functions:

(a) $f(x) = x^4 - 3x^3 + 16x$ (b) $h(x) = \sqrt{x^2+1}$ (c) $f(x) = \dfrac{x}{1-x}$

(d) $y = (1-x^2)^{3/4}$ (e) $f(x) = e^{2x-\sin(5x)}$ (f) $g(x) = \operatorname{cosec}(x)$

Q2. nth Derivatives. For the following functions evaluate the first few derivatives. Deduce the pattern and write down an expression for the nth derivative $f^{(n)}(x)$:

(a) $f(x) = x^n$ (b) $f(x) = \dfrac{1}{3x^3}$

Q3. Related Derivatives. Assuming $g(x)$ is twice differentiable, that is, we can write down $g'(x)$ and $g''(x)$, write down expressions for $f''(x)$ if:

(a) $f(x) = xg(x^2)$ (b) $f(x) = g(\sqrt{x})$

Graph sketching

Q4. Graph sketching. For each of the following functions:

(a) $f(x) = \dfrac{1+x^2}{1-x^2}$ (b) $f(x) = \dfrac{x^3}{x^2+1}$ (c) $f(x) = x\sqrt{9-x^2}$

(d)(*) $f(x) = e^{1/x}$ (e) $f(x) = \dfrac{1}{x^2-1}$

Determine where possible

- domain of f
- asymptotes of f
- x and y intercepts of the graph $y = f(x)$
- local extrema of f
- intervals over which f is increasing
- intervals over which f is decreasing
- intervals over which f is concave up
- intervals over which f is concave down
- points of inflection of f

Use these results to sketch the graph of $y = f(x)$.

[Hint for (d): Consider left and right limits as x approaches 0.]

Implicit Differentiation

Q5. Implicit Derivatives. Find $\dfrac{dy}{dx}$ by implicit differentiation:

(a) $x^2 - xy + y^3 = 8$

(b) $4\cos x \sin y = 1$

(c) $x \log_e y + \sqrt{y} = \log_e x$

(d) $x^2 y = e^{xy}$

(e) $\cos(x - y) = xe^x$

(f) $\log_e \frac{x}{y} = \cot(xy)$

Q6. Second Derivative. If $x^4 + y^4 = 16$, use the following steps to find y'':
(a) Use implicit differentiation to find y'

(b) Use the quotient rule to differentiate the expression for y'

from part (a). Express your answer in terms of x and y only.
(c) Use the fact that x and y must satisfy the original equation $x^4 + y^4 = 16$ to simplify your answer to part (b) to $y'' = -48\dfrac{x^2}{y^7}$.

Q7. More Implicit Derivatives Suppose y is given implicitly as a function of x by the equation

$$x^2 + axy^2 + by^3 = 1.$$

Given that this curve passes through the point $(3, 2)$ and that the tangent line that touches the curve at this point has a gradient of -1, find a and b.

Inverse Trigonometric Functions

Q8. Derivatives. Find the derivative with respect to x of each of the following expressions:

(a) $\arcsin\left(\frac{x}{5}\right)$

(b) $\arccos\left(\frac{x}{4}\right)$

(c) $\arctan\left(\frac{x}{\sqrt{7}}\right)$

(d) $\arccos(4x)$

(e) $\arctan(7x)$

(f) $\arcsin(2x)$

Q9. More derivatives. Find $f'(x)$ if $f(x)$ is equal to:

(a) $\arccos\left(\frac{7x}{4}\right)$

(b) $\arcsin\left(\frac{8x}{5}\right)$

(c) $\arctan\left(\frac{7x}{2}\right)$

(d) $\arccos(4x - 3)$

(e) $\arcsin\left(\frac{3-4x}{5}\right)$

(f) $\arctan\left(\frac{5x-3}{4}\right)$

Q10. Even more derivatives. Find $f'(x)$ if $f(x)$ is equal to:

(a) $\arccos(x^3)$

(b) $\arctan(e^x + 5)$

(c) $\arcsin(\sqrt[3]{x})$

(d) $\arccos(2\log(x))$

(e) $\arctan(\cos(3x) + 1)$

(f) $\arcsin\left(\dfrac{2x^2 + 1}{3}\right)$

Applications of Differentiation

Q11. Flying High. The height h, in metres, of a kite above the ground t minutes after it is let off is given by

$$h(t) = 12t - 2t^2 \ .$$

(a) Find when the kite reaches its maximum height, and the maximum height reached.

(b) Find the time when the kite falls to the ground.

Q12. Flying Higher. In a second attempt, the kite now flies at height

$$h(t) = \frac{1}{6}t^3 - \frac{9}{4}t^2 + 9t \qquad \text{for } 0 \le t \le 10.$$

(a) What is the maximum height reached by the kite in the first 5 minutes?

(b) What is the maximum height reached by the kite in the whole 10 minutes?

Q13. Cylinder. The radius, r cm, and height, h cm, of a solid circular cylinder vary in such a way that the volume of the cylinder is always 250π cm^3. (a) Draw the cylinder and write down expressions for its total surface area, A cm^2, and its volume, V cm^3, in terms of r and h.

(b) Show that the total surface area can be rewritten as:

$$A = 2\pi r^2 + \frac{500\pi}{r} \ .$$

(c) Find the minimum total surface area of the cylinder.

Q14. Rectangle. Find the maximum possible area of a rectangle which has its base on the x-axis and its upper vertices on the semi-circle $y = \sqrt{9 - x^2}$, $-3 \le x \le 3$.

Q15. Manufacturing. The sales revenue (in dollars) that a manufacturer receives for selling x units of a certain product can be approximated by the function

$$R(x) = 900 \log\left(1 + \frac{x}{300}\right) \ .$$

However, each unit costs the manufacturer one dollar to produce, and the initial cost of adjusting the machinery for the production is \$200, so the total cost of production (in dollars) of x units is

$$C(x) = 200 + x \ .$$

(a) Write down the profit, $P(x)$ dollars, obtained by the production and sales of x units.

(b) Hence find the number of units which should be produced and sold for maximum profit, and calculate this maximum profit.

Q16. Rock n Roll. A travelling rock band is having a trailer made to carry their equipment. The roadies say the trailer must have a square base, an open top, and a volume of 32m^3. Find the dimensions (length of base side and the height) for the trailer to be made with the least amount of sheet metal.

Differentiating Parametric Functions

Q17. Velocity, speed and acceleration. For each of the following parametric curves $\mathbf{r} : \mathbb{R} \longrightarrow \mathbb{R}^2$, find formulas for the velocity, speed and acceleration at time t.

(a) $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(3t)\mathbf{j}$.

(b) $\mathbf{r}(t) = t^5\mathbf{i} + t^7\mathbf{j}$.

(c) $\mathbf{r}(t) = e^{-t^2}\mathbf{i} + t^3\mathbf{j}$.

(d) $\mathbf{r}(t) = e^{-t^2}\mathbf{i} + e^{-t^3}\mathbf{j}$.

Q18. Cusps and smoothness. Decide whether each of the following parametric curves $\mathbf{r} : \mathbb{R} \longrightarrow \mathbb{R}^2$ is smooth. For those that are not smooth, find any cusps.

(a) $\mathbf{r}(t) = e^{-t^2}\mathbf{i} + t^3\mathbf{j}$.

(b) $\mathbf{r}(t) = t^5\mathbf{i} + t^7\mathbf{j}$

(c) $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(3t)\mathbf{j}$.

(d) $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(t)\mathbf{j}$.

(e) $\mathbf{r}(t) = (2\cos(t) + \cos(2t))\mathbf{i} + (2\sin(t) - \sin(2t))\mathbf{j}$.

Q19. Variation in speed. For each of the following parametric curves, find the set of t values for which the speed is (i) increasing, (ii) decreasing and (iii) stationary.

(a) $\mathbf{r}(t) = \frac{1}{2}t^2\mathbf{i} - (\frac{1}{3}t^3 - t)\mathbf{j}$.

(b) $\mathbf{r}(t) = a\cos(t)\mathbf{i} + b\sin(t)\mathbf{j}$ where $a > b > 0$.

Q20. Putting it all together. Suppose that the motion of two particles are given by the parametric curves $\mathbf{r}_1 : \mathbb{R} \longrightarrow \mathbb{R}$ and $\mathbf{r}_2 : \mathbb{R} \longrightarrow \mathbb{R}$, where:

$$\mathbf{r}_1(t) = \left(t^3 - t^2\right)\mathbf{i} + \left(t^3 + t^2\right)\mathbf{j}$$
$$\mathbf{r}_2(t) = \left(t^2 + 3t\right)\mathbf{i} + (t+1)\mathbf{j}$$

(a) Find the *time* and *position* where the particles collide.

(b) Find the intervals on which the speed of the particle given by $\mathbf{r}_1$ is:

(i) increasing. (ii) decreasing.

(c) Decide whether the speed of each of the particle is increasing or decreasing the time of collision.

A1. First and Second Derivatives.

(a) $f'(x) = 4x^3 - 9x^2 + 16$, $f''(x) = 12x^2 - 18x$ (b) $h'(x) = x/(\sqrt{x^2+1})$, $h''(x) = 1/(x^2+1)^{3/2}$

(c) $y' = 1/(1-x)^2$, $y'' = 2/(1-x)^3$ (d) $y' = -\frac{3}{2}x(1-x^2)^{-1/4}$, $y'' = \frac{3}{4}(1-x^2)^{-5/4}(x^2-2)$

(e) $f'(x) = (2 - 5\cos(5x))e^{2x-\sin(5x)}$, $f''(x) = (25\sin(5x) + (2-5\cos(5x))^2)e^{2x-\sin(5x)}$

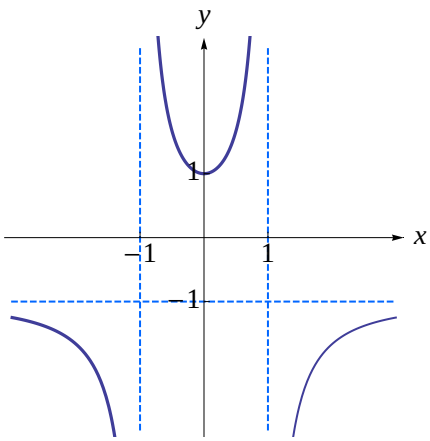
(f) $g'(x) = -\operatorname{cosec}(x)\cot(x)$, $g''(x) = \operatorname{cosec}^3(x) + \operatorname{cosec}(x)\cot^2(x)$

A2. nth Derivatives. (a) $n! = n \times (n-1) \times (n-2) \times (n-3)...$ (b) $(-1)^n(n+2)!/(6x^{n+3})$

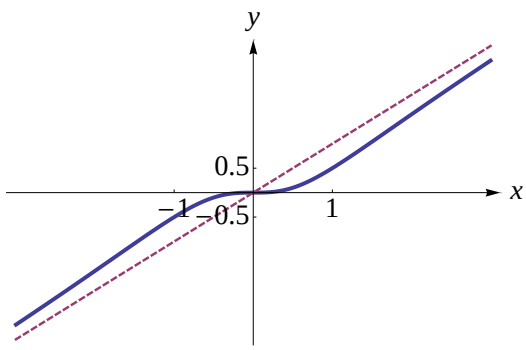
A3. Related Derivatives. (a) $f''(x) = 6xg'(x^2) + 4x^3g''(x^2)$ (b) $f''(x) = \frac{\sqrt{x}g''(\sqrt{x}) - g'(\sqrt{x})}{4x\sqrt{x}}$

A4. Graph sketching.

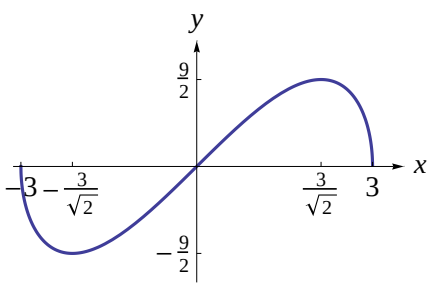
(a) Domain $\mathbb{R}$ except ± 1. y intercept at $y = 1$, no x intercept. Local minimum at $(0, 1)$. Increasing for $0 < x < 1$ and $x > 1$, decreasing for $x < -1$ and $-1 < x < 0$. Concave down for $x < -1$ and $x > 1$, concave up for $-1 < x < 1$. No points of inflection. Vertical asymptotes at $x = \pm 1$, horizontal asymptote at $y = -1$.



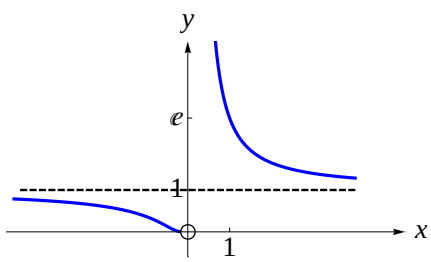
(b) Domain $\mathbb{R}$. Oblique asymptote $y = x$. x and y-intercept at the origin $(0, 0)$. Stationary point at $x = 0$. Increasing for all x. Concave up for $x < -\sqrt{3}$ and $0 < x < \sqrt{3}$. Concave down for $-\sqrt{3} < x < 0$ and $x > \sqrt{3}$. Points of inflection at $x = -\sqrt{3}, 0, \sqrt{3}$.



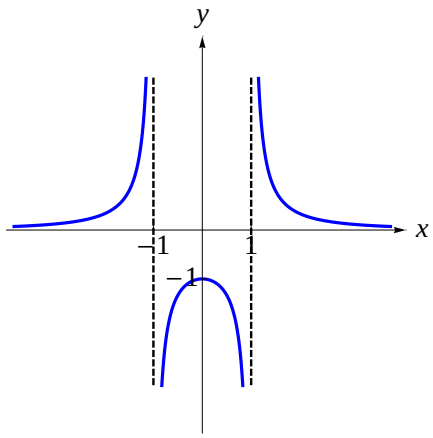
(c) Domain $-3 \le x \le 3$. Intercepts are (-3,0), (0,0), (3,0). Local minimum at $(-3\sqrt{2}/2, -9/2)$. Local maximum at $(3\sqrt{2}/2, 9/2)$. Decreasing for $x < -3\sqrt{2}/2$ and $x > 3\sqrt{2}/2$. Increasing on $-3\sqrt{2}/2 < x < 3\sqrt{2}/2$. Concave up for $x < 0$, concave down for $x > 0$. Point of inflection at $x = 0$.



(d) Domain $\mathbb{R}$ except 0. Vertical asymptote at $x = 0$, horizontal asymptote at $y = 1$. No x or y intercepts. No stationary points. Decreasing for all $x \in \mathbb{R}\backslash\{0\}$. Concave up for $-\frac{1}{2} < x < 0$ and $x > 0$, concave down for $x < -\frac{1}{2}$. Point of inflection at $x = -\frac{1}{2}$. As $x \longrightarrow 0$ from above, $f(x) \longrightarrow \infty$, while as $x \longrightarrow 0$ from below, $f(x) \longrightarrow 0$.



(e) Domain $\mathbb{R}$ except $\{-1, 1\}$. Vertical asymptotes at $x = -1, 1$, horizontal asymptote at $y = 0$. No x intercept; y intercept $(0, -1)$. Local maximum at $(0, -1)$. Increasing on $(-\infty, -1) \cup (-1, 0)$. Decreasing on $(0, 1) \cup (1, \infty)$. No points of inflection. Concave up on $(-\infty, -1) \cup (1, \infty)$, concave down on $(-1, 1)$.



A5. Implicit Derivatives.

(a) $\dfrac{y - 2x}{3y^2 - x}$ (b) $\tan x \tan y$ (c) $\dfrac{2y - 2xy \log_e y}{2x^2 + x\sqrt{y}}$ (d) $\dfrac{ye^{xy} - 2xy}{x^2 - xe^{xy}}$

(e) $1 + \dfrac{e^x(1 + x)}{\sin(x - y)}$ (f) $\dfrac{y + y^2 x \operatorname{cosec}^2 xy}{x - x^2 y \operatorname{cosec}^2 xy}$

A6. Second Derivative. (a) $y' = \dfrac{-x^3}{y^3}$ (b) $y'' = \dfrac{-3x^2}{y^3} - \dfrac{3x^6}{y^7}$

A7. More Implicit Derivatives $a = -9/5$ and $b = 17/10$.

A8. Derivatives.

(a) $\dfrac{1}{\sqrt{25 - x^2}}$ (b) $-\dfrac{1}{\sqrt{16 - x^2}}$ (c) $\dfrac{\sqrt{7}}{7 + x^2}$

(d) $-\dfrac{4}{\sqrt{1 - 16x^2}}$ (e) $\dfrac{7}{1 + 49x^2}$ (f) $\dfrac{2}{\sqrt{1 - 4x^2}}$

A9. More derivatives.

(a) $-\dfrac{7}{\sqrt{16 - 49x^2}}$ (b) $\dfrac{8}{\sqrt{25 - 64x^2}}$ (c) $\dfrac{14}{4 + 49x^2}$

(d) $-\dfrac{4}{\sqrt{1 - (4x - 3)^2}}$ (e) $-\dfrac{4}{\sqrt{25 - (3 - 4x)^2}}$ (f) $\dfrac{20}{16 + (5x - 3)^2}$

A10. Even more derivatives.

(a) $-\dfrac{3x^2}{\sqrt{1 - x^6}}$ (b) $\dfrac{e^x}{1 + (e^x + 5)^2}$ (c) $\dfrac{1}{3x^{2/3}\sqrt{1 - x^{2/3}}}$

(d) $-\dfrac{2}{x\sqrt{1 - 4(\log(x))^2}}$ (e) $-\dfrac{3\sin(3x)}{1 + (\cos(3x) + 1)^2}$ (f) $\dfrac{4x}{\sqrt{9 - (2x^2 + 1)^2}}$

A11. Flying High. (a) $h = 18$ at $t = 3$ (b) $t = 6$

A12. Flying Higher. (a) $h = 11\frac{1}{4}$ at $t = 3$ (b) $h = 31\frac{2}{3}$ at $t = 10$

A13. Cylinder.

(a) $A = 2\pi r^2 + 2\pi rh, \;\; V = \pi r^2 h$

(b) Proof required.

(c) 150π cm^2

A14. Rectangle. Maximum area is 9, when one vertex is at $(\frac{3}{\sqrt{2}}, 0)$.

A15. Manufacturing.

(a) $P(x) = 900\log(1 + \frac{x}{300}) - (200 + x)$ (b) $x = 600$ gives maximum profit of \$188.75.

A16. Rock n Roll. Base: 4m by 4m, Height: 2m.

A17. Velocity, speed and acceleration.

(a) Velocity: $\mathbf{r}'(t) = -\sin(t)\mathbf{i} + 3\cos(3t)\mathbf{j}$. Speed $\|\mathbf{r}'(t)\| = \sqrt{\sin^2(t) + 9\cos^2(3t)}$.

Acceleration: $\mathbf{r}''(t) = -\cos(t)\mathbf{i} - 9\sin(3t)\mathbf{j}$.

(b) Velocity: $\mathbf{r}'(t) = 5t^4\mathbf{i} + 7t^6\mathbf{j}$. Speed $\|\mathbf{r}'(t)\| = t^4\sqrt{25 + 49t^4}$.

Acceleration: $\mathbf{r}''(t) = 20t^3\mathbf{i} + 42t^5\mathbf{j}$.

(c) Velocity: $\mathbf{r}'(t) = -2te^{-t^2}\mathbf{i} + 3t^2\mathbf{j}$. Speed $\|\mathbf{r}'(t)\| = |t|\sqrt{4e^{-2t^2} + 9t^2}$.

Acceleration: $\mathbf{r}''(t) = 2e^{-t^2}(2t^2 - 1)\mathbf{i} + 6t\mathbf{j}$.

(d) Velocity: $\mathbf{r}'(t) = -2te^{-t^2}\mathbf{i} - 3t^2e^{-t^3}\mathbf{j}$. Speed $\|\mathbf{r}'(t)\| = |t|\sqrt{4e^{-2t^2} + 9t^2e^{-2t^3}}$.

Acceleration: $\mathbf{r}''(t) = 2e^{-t^2}(2t^2 - 1)\mathbf{i} + 3te^{-t^3}(3t^3 - 2)\mathbf{j}$.

A18. Cusps and smoothness.

(a) If $\mathbf{r}'(t) = -2te^{-t^2}\mathbf{i} + 3t^2\mathbf{j} = \mathbf{0}$, then $t = 0$. Therefore $x'(t) = -2te^{-t^2}$ changes sign at $t = 0$ And so there is a cusp at $\mathbf{r}(0) = \mathbf{i}$.

(b) If $\mathbf{r}'(t) = 5t^4\mathbf{i} + 7t^6\mathbf{j} = \mathbf{0}$, then $t = 0$ Neither $x'(t) = 5t^4$ nor $x'(t) = 7t^6$ changes sign at $t = 0$ And so there are no cusps. The curve is smooth.

(c) If $\mathbf{r}'(t) = -\sin(t)\mathbf{i} + 3\cos(3t)\mathbf{j} = \mathbf{0}$, then $\sin(t) = \cos(3t) = 0$. This implies $t \in \{k\pi \mid k \in \mathbb{Z}\} \cap \{\frac{\pi}{6} + \frac{k\pi}{3} \mid k \in \mathbb{Z}\}$. But $\{k\pi \mid k \in \mathbb{Z}\} \cap \{\frac{\pi}{6} + \frac{k\pi}{3} \mid k \in \mathbb{Z}\} = \emptyset$, so there are no cusps. The curve is smooth.

(d) If $\mathbf{r}'(t) = -4\sin(t)\cos(t)\mathbf{i} + \cos(t)\mathbf{j} = \mathbf{0}$, then $\cos(t) = 0$. This implies $t \in \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$. And so $x'(t) = \cos(t)$ changes sign at each of these t values. Therefore there are cusps at $\mathbf{r}(\frac{\pi}{2}) = -\mathbf{i} + \mathbf{j}$ and $\mathbf{r}(\frac{3\pi}{2}) = -\mathbf{i} - \mathbf{j}$.

(e) If $\mathbf{r}'(t) = -2(\sin(t) + \sin(2t))\mathbf{i} + 2(\cos(t) - \cos(2t))\mathbf{j} = \mathbf{0}$, then $\sin(t) = -\sin(2t)$ and $\cos(t) = \cos(2t)$.

Using the identities $\sin(2t) = 2\sin(t)\cos(t)$ and $\cos(2t) = 2\cos^2(t) - 1$ we can solve $\sin(t) = -\sin(2t)$ and $\cos(t) = \cos(2t)$ to obtain ($\sin(t) = 0$ or $\cos(t) = -\frac{1}{2}$) and $\cos(t) \in \{-\frac{1}{2}, 1\}$ respectively.

These conditions hold for all $t \in \{2k\pi \mid k \in \mathbb{Z}\} \cup \{\frac{2\pi}{3} + 2k\pi \mid k \in \mathbb{Z}\} \cup \{\frac{-2\pi}{3} + 2k\pi \mid k \in \mathbb{Z}\}$, giving cusps at $3\mathbf{i}, -\frac{3}{2}\mathbf{i} + \frac{3\sqrt{3}}{2}\mathbf{j}$ and $-\frac{3}{2}\mathbf{i} - \frac{3\sqrt{3}}{2}\mathbf{j}$.

A19. Variation in speed.

(a) $\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 2t^3 - t = t(2t^2 - 1)$. From this we can show that speed is increasing on $(-\frac{1}{\sqrt{2}}, 0) \cup (\frac{1}{\sqrt{2}}, \infty)$, decreasing on $(-\infty, -\frac{1}{\sqrt{2}}) \cup (0, \frac{1}{\sqrt{2}})$ and stationary at $0, \pm\frac{1}{\sqrt{2}}$.

(b) $\mathbf{r}'(t) \cdot \mathbf{r}''(t) = a^2 \sin(t)\cos(t) - b^2 \sin(t)\cos(t) = \frac{1}{2}(a^2 - b^2)\sin(2t))$ and we are given $a > b$ so the sign of $\mathbf{r}'(t) \cdot \mathbf{r}''(t)$ is given by the sign of $\sin(2t) = 0$. Thus, speed is increasing on each interval $(k\pi, k\pi + \frac{\pi}{2})$, decreasing on each interval $(k\pi + \frac{\pi}{2}, k\pi + \pi)$ and stationary at each point of $\{\frac{k\pi}{2} \mid k \in \mathbb{Z}\}$.

A20. Putting it all together.

(a) Particles collide at $-2i$ when $t = -1$.

(b) (i) increasing on $(0, \infty)$ (ii) decreasing on $(-\infty, 0)$

(c) Speed of particle given by $\mathbf{r}_1(t)$ decreasing when $t = -1$. Speed of particle given by $\mathbf{r}_2(t)$ is increasing when $t = -1$.

Useful Formulae

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1$$

Compound angle formulae

$$\begin{aligned}
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\tan(x+y) &= \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}
\end{aligned}$$

Derivatives of inverse trigonometric functions

$$\begin{aligned}
\arcsin'(x) &= \frac{1}{\sqrt{1-x^2}} \\
\arccos'(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\arctan'(x) &= \frac{1}{1+x^2}
\end{aligned}$$

Antiderivatives from inverse trigonometric functions

$$\begin{aligned}
\int \frac{1}{\sqrt{s^2-x^2}}\,dx &= \arcsin\left(\frac{x}{s}\right) + C \\
\int \frac{-1}{\sqrt{s^2-x^2}}\,dx &= \arccos\left(\frac{x}{s}\right) + C \\
\int \frac{1}{s^2+x^2}\,dx &= \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C
\end{aligned}$$

where s is a *positive* constant, and C is an arbitrary constant of integration.

Complex exponential formulae

$$\begin{aligned}
e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\
\cos(\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \\
\sin(\theta) &= \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)
\end{aligned}$$

Vector projections

- The *vector projection* of $\mathbf{v}$ onto $\mathbf{u}$ is $\mathbf{v}_{\parallel} = (\hat{\mathbf{u}} \cdot \mathbf{v})\hat{\mathbf{u}} = k\mathbf{u}$, where $k \in \mathbb{R}$ is the unique solution of $\mathbf{u}\cdot(\mathbf{v}-k\mathbf{u}) = 0$.
- The *vector component of* $\mathbf{v}$ *perpendicular to* $\mathbf{u}$ is $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$.

Complex roots

The n-th roots of $w = se^{i\phi}$ are $s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\phi+2k\pi))}$ for $k = 0, 1, \ldots, n-1$.

Changes in speed

Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the speed function $\|\mathbf{r}'(t)\|$ is decreasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) < 0$ and increasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) > 0$.